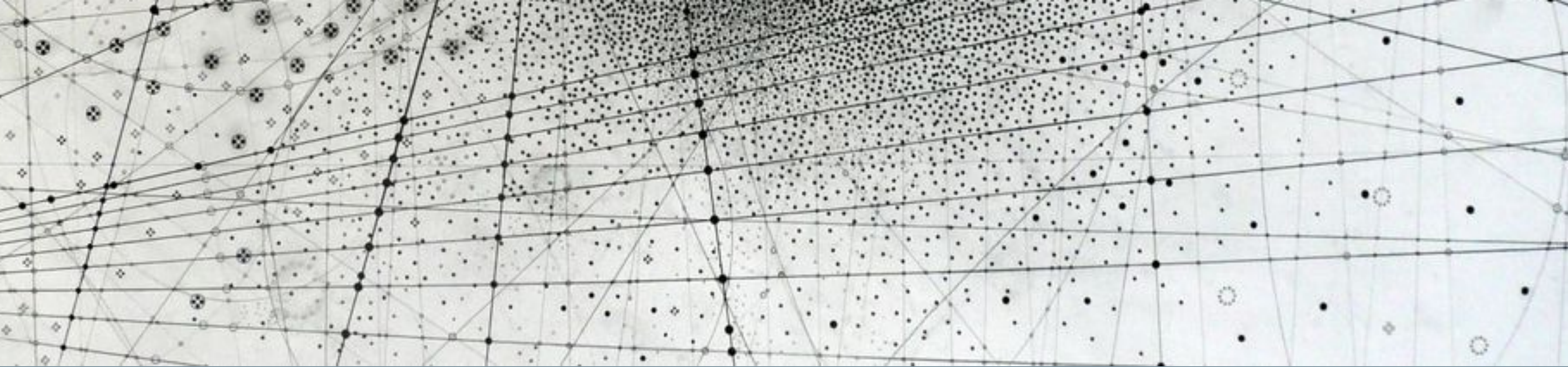


Research Methods in Machine Learning & Physics

Quick recap

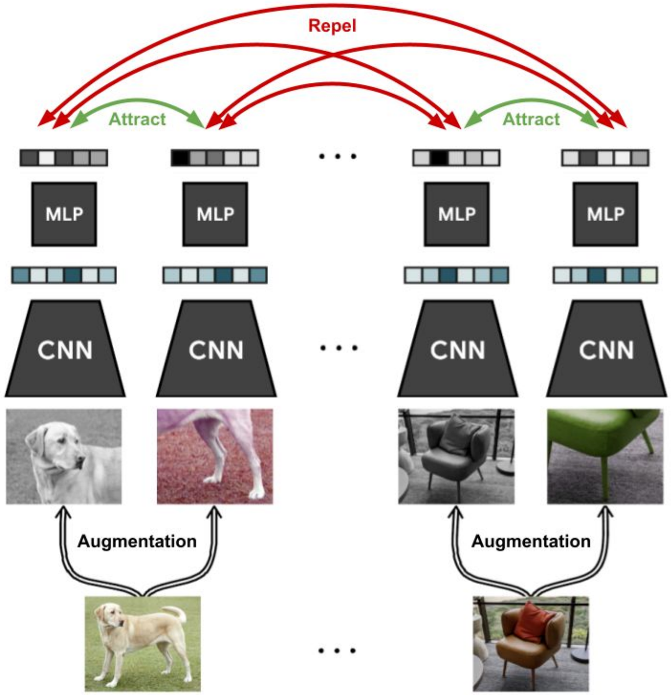
Self-supervised learning (SSL)

SSL = creating labels from the data itself in a semi-automatic way

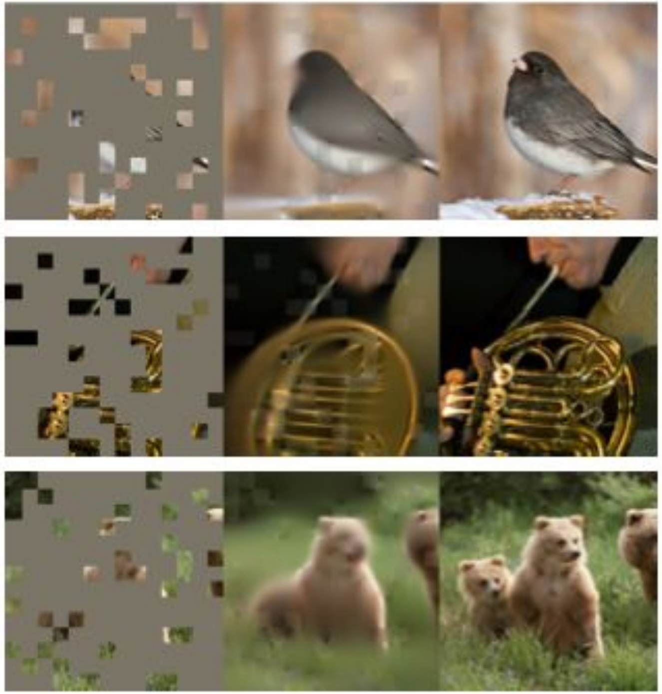
In other words, predicting part of the data using other parts

Contrastive vs. non-contrastive SSL

Contrastive SSL: compare 2 augmentations



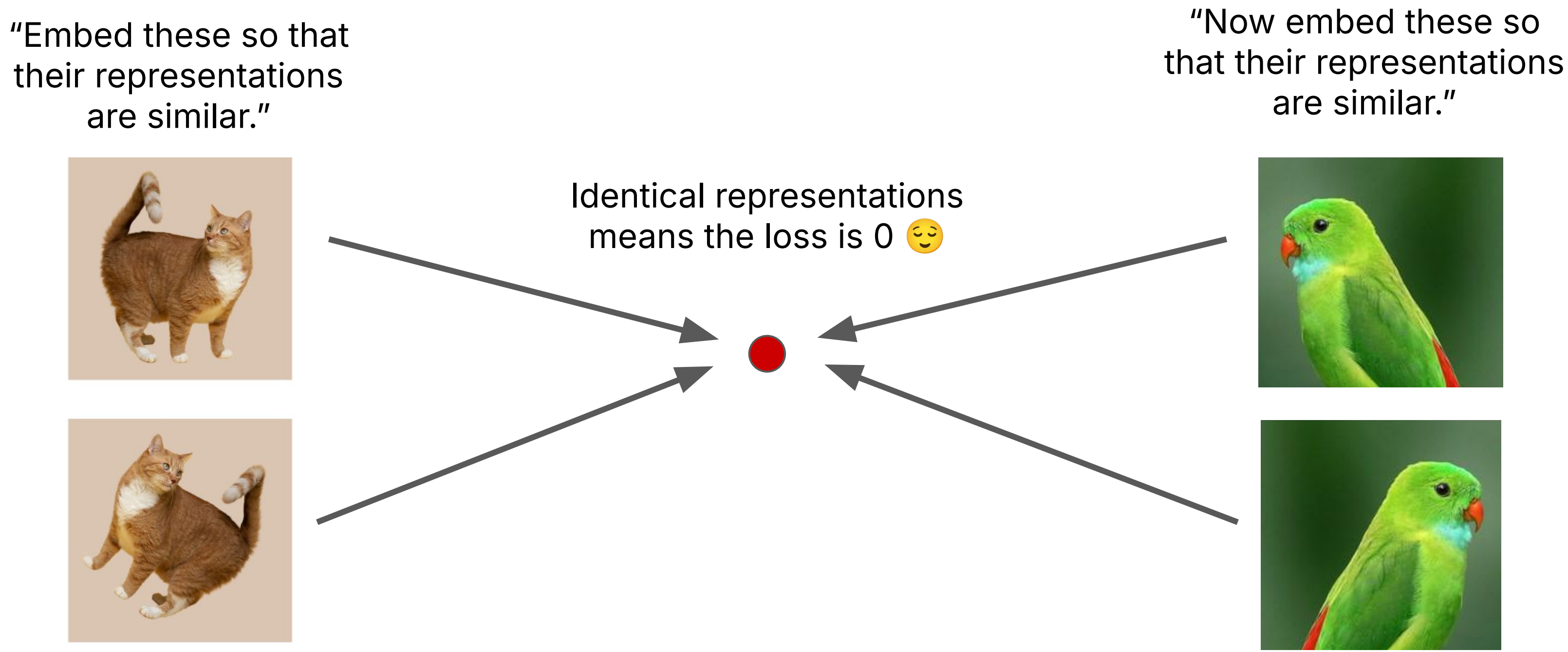
Generative SSL: randomly mask, then predict



Non-contrastive SSL, Part 2

Moving toward a unified picture of SSL

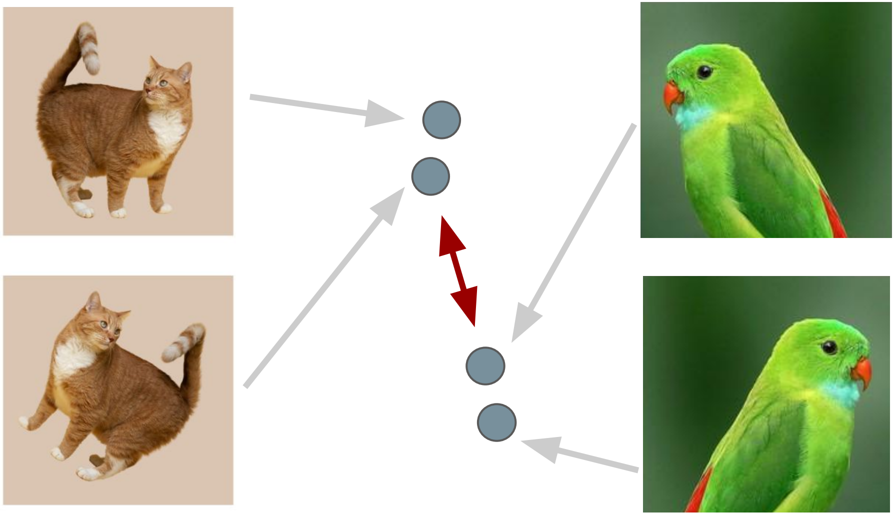
All SSL methods are designed to avoid the following scenario:



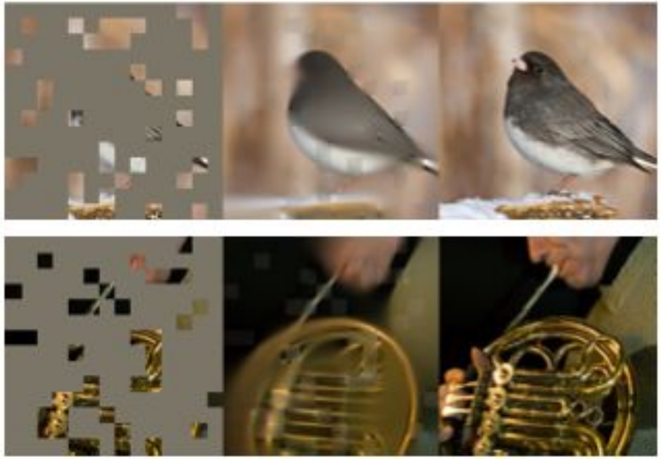
Moving toward a unified picture of SSL

How can we avoid learning a trivial solution? ⟶ Need to **enforce some variation!**

Contrastive methods use **negative pairs** to push apart various representations:



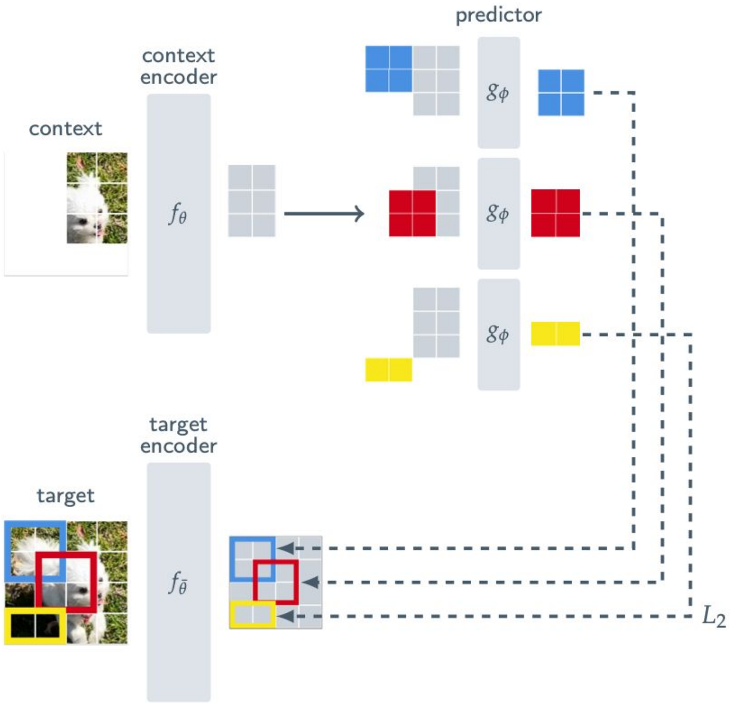
Generative methods use **masking:**

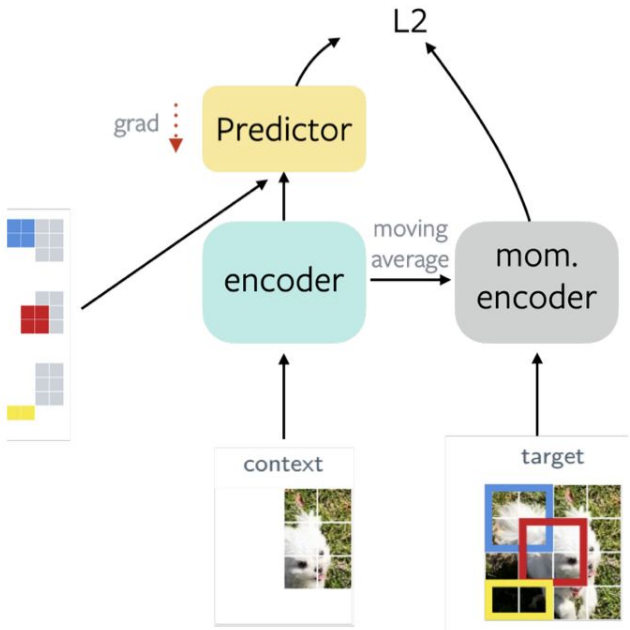


Note: only possible if you have a discrete input space!

Masked modeling in latent space:
JEPA (Joint-Embedding Predictive Architecture)

Masking can also be done **in the embedding space** instead of the "pixel" (i.e. input) space:





Other non-contrastive SSL methods

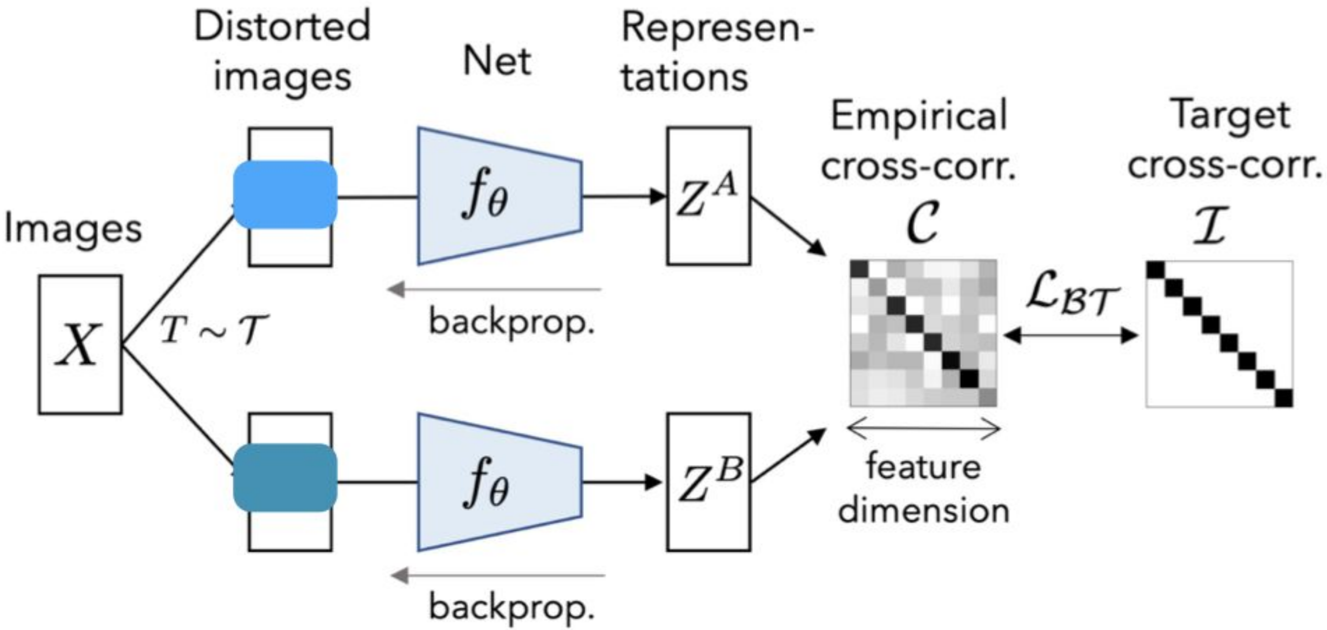
There are a few other strategies for enforcing varied outputs:

Apply statistical constraints (“redundancy reduction”)
- VICReg
- Barlow Twins

Use an asymmetric architecture (“self-distillation”)
- BYOL
- DINO

Apply statistical constraints: Barlow Twins

Enforce that the empirical cross-correlation is close to the identity matrix: **reduce redundant info!**



Apply statistical constraints: VICReg

Adds penalties to **V**ariance, **I**nvariance, and **C**ovariance terms

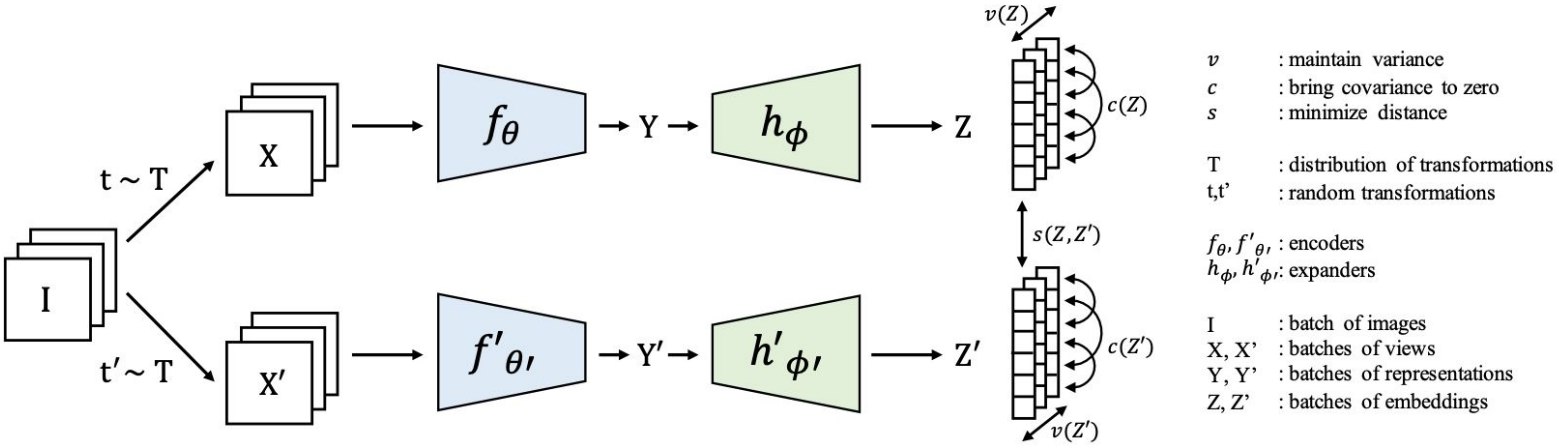


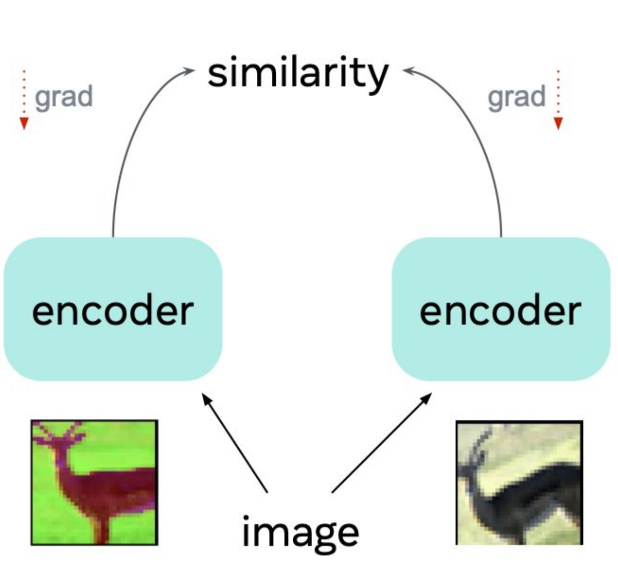
Figure 6: **VICReg**: penalizes variance, invariance, and co-variance terms to learn representations from unlabeled data.

Use an asymmetric architecture:

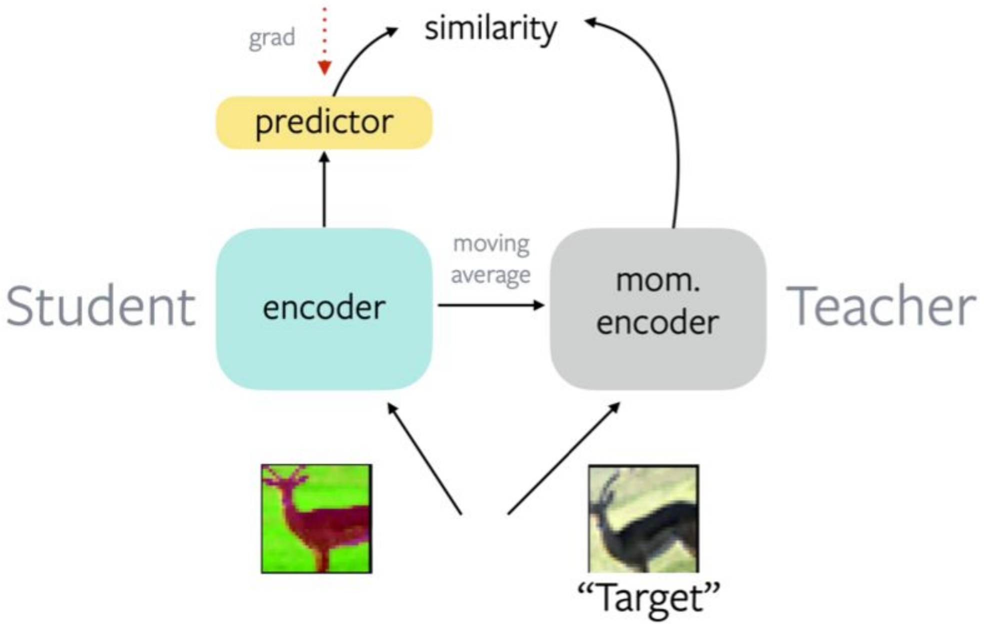
BYOL (Bootstrap Your Own Latent)

Instead of directly enforcing similarity...

$$f_\theta(I) = f_\theta(\text{augment}(I))$$



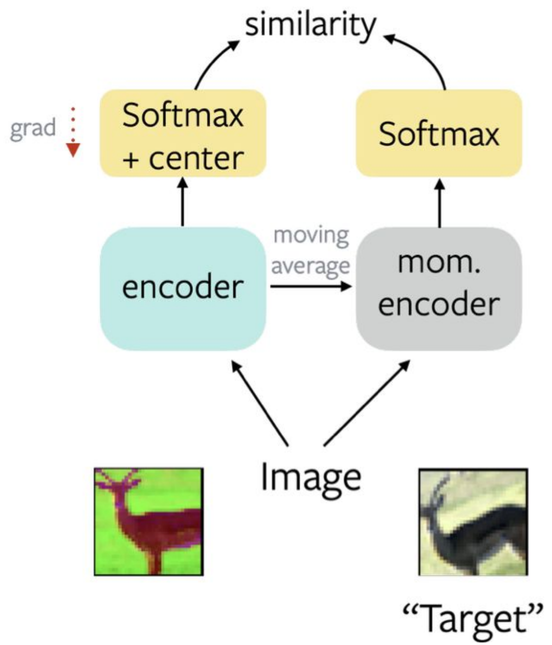
$$f_\theta^{\text{student}}(I) = f_\theta^{\text{teacher}}(\text{augment}(I))$$



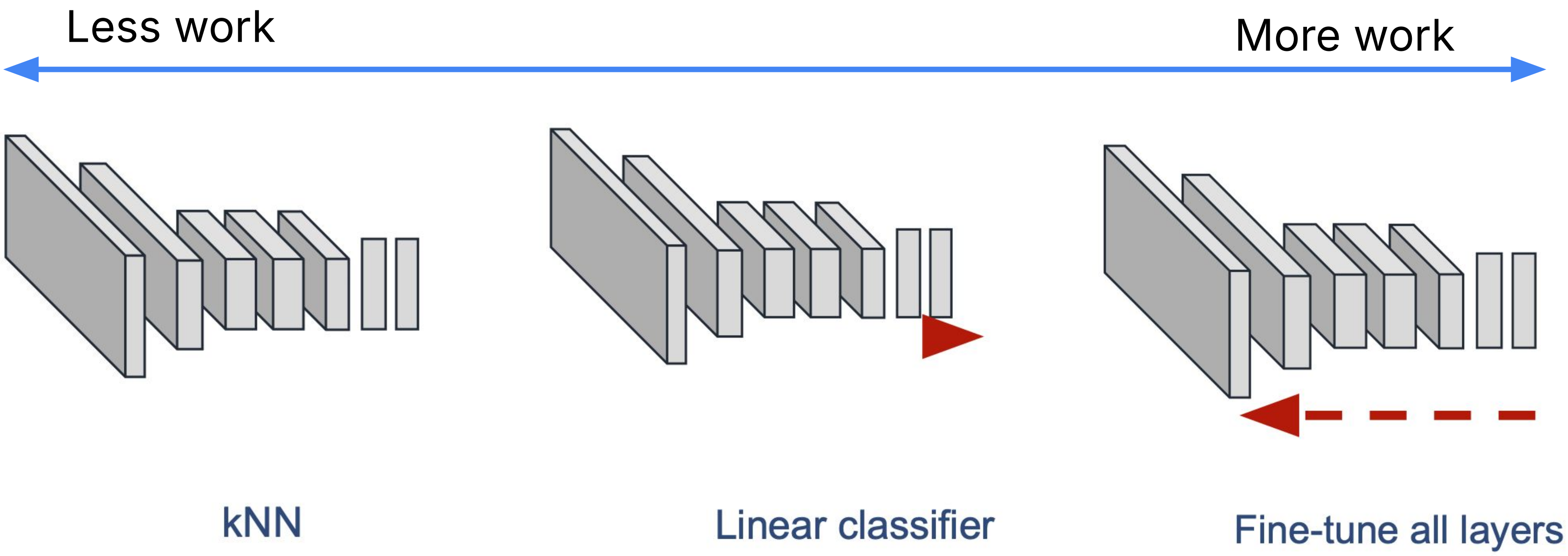
Use an asymmetric architecture:
DINO (Self-DIstillation with NO labels)

Avoiding collapse. Several self-supervised methods differ by the operation used to avoid collapse, either through contrastive loss [73], clustering constraints [8, 10], predictor [30] or batch normalizations [30, 58]. While our framework can be stabilized with multiple normalizations [10], it can also work with only a centering and sharpening of the momentum teacher outputs to avoid model collapse. As shown experimentally in Section 5.3, centering prevents one dimension to dominate but encourages collapse to the uniform distribution, while the sharpening has the opposite effect. Applying both operations balances their effects which is sufficient to avoid collapse in presence of a momentum teacher. Choosing this method to avoid collapse trades stability for less dependence over the batch: the centering operation only depends on first-order batch statistics and can be interpreted as adding a bias term c to the teacher:

$$f_\theta^{\text{student}}(I) = f_\theta^{\text{teacher}}(\text{augment}(I))$$

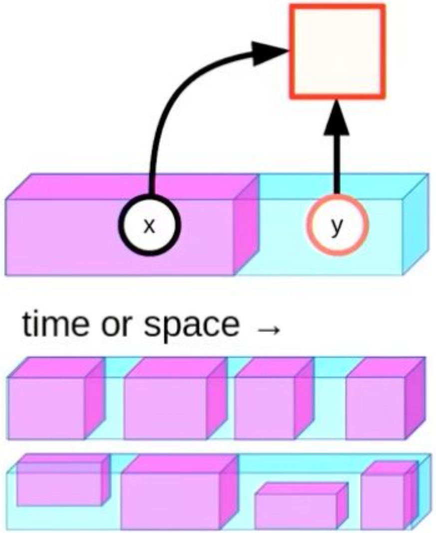


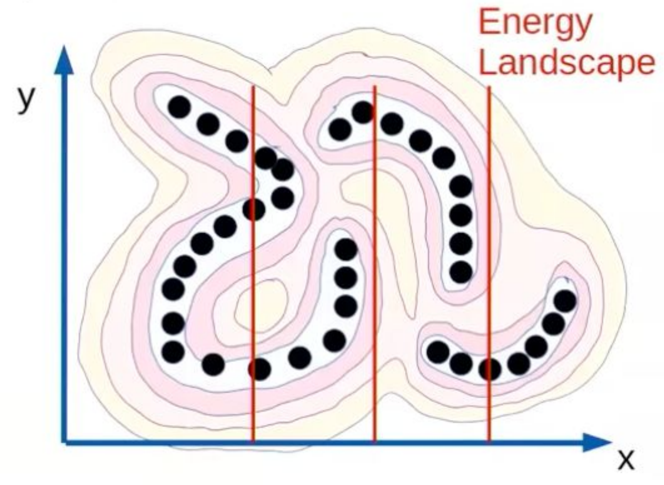
How to evaluate a model trained via SSL?



Unified theories of self-supervised learning

- **The only way to formalize & understand all model types**
  - Gives low energy to compatible pairs of x and y
  - Gives higher energy to incompatible pairs



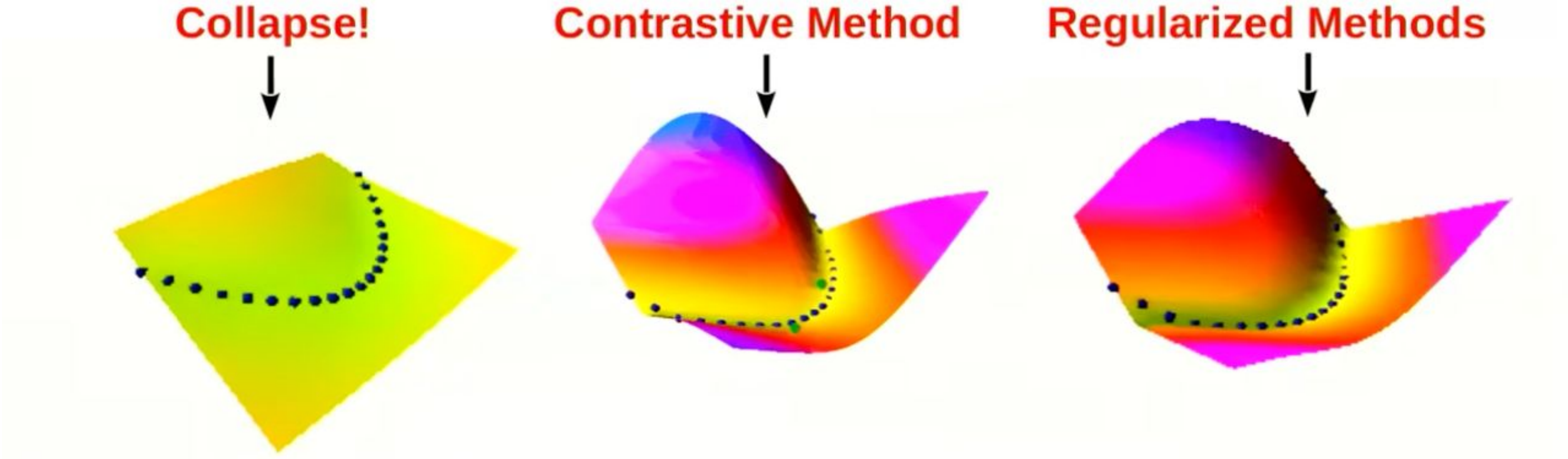


$$\check{y} = \mathrm{argmin}_y F(x, y)$$

https://www.youtube.com/watch?v=yUmDRxV0krg

Unified theories of self-supervised learning

- A flexible energy surface can take any shape.
- We need a loss function that shapes the energy surface so that:
  - Data points have low energies
  - Points outside the regions of high data density have higher energies.



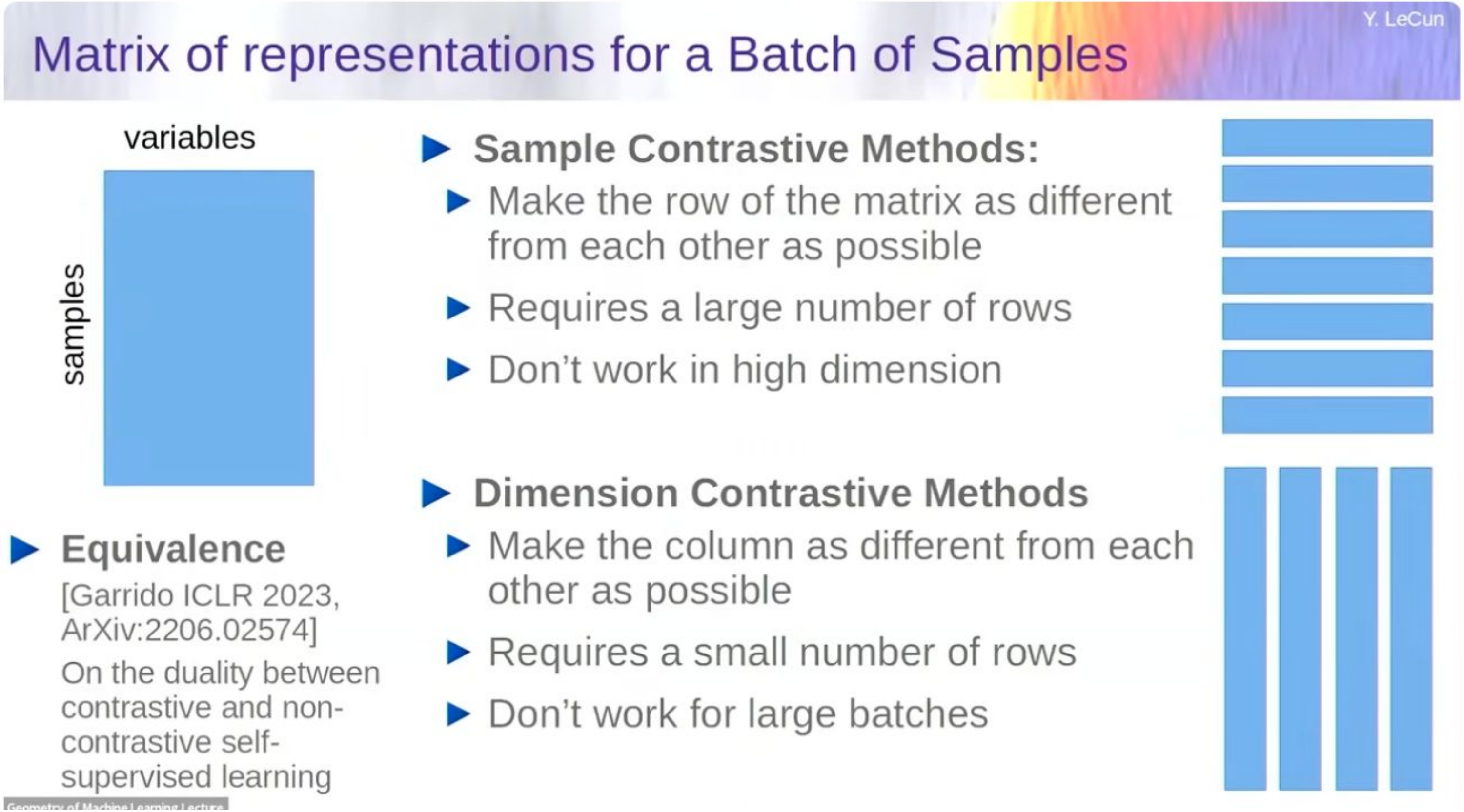
https://www.youtube.com/watch?v=yUmDRxV0krg

Unified theories of self-supervised learning

6 CONCLUSION

Through an analysis of their criteria, we were able to show that sample-contrastive and dimension-contrastive methods have learning objectives that are closely related, as they are effectively minimizing criteria that are equivalent up to row and column normalization of the embedding matrix. This suggests a certain duality in the behavior of such methods, which we studied empirically. Through the lens of variations of VICReg, we were able to study popular design choices in self-supervised loss functions and show their lack of impact on performance, significantly improving the robustness to embedding dimension of VICReg along the way. Motivated by our theoretical findings, we performed ample hyperparameter tuning on SimCLR and were able to close its performance gap with VICReg. We also showed that the normalization strategy does not play an important role in performance. This further reinforces the similarities between methods as predicted by our theoretical results. We expect that our results will help extend theoretical works in self-supervised learning to a wider family of methods, as well as help analyses by deriving criteria that are easier to work with. We also expect that our findings will help alleviate preconceived ideas on contrastive and non-contrastive learning. If one thing must be remembered from this work, it is that *dimension-contrastive and sample-contrastive methods are two sides of the same coin.* Finally, perhaps the most important message of this work is to show that different SOTA SSL methods can be unified.

Unified theories of self-supervised learning



Resources

2023 ICML tutorial on SSL: https://icml.cc/virtual/2023/tutorial/21552

A Cookbook of Self-Supervised Learning:
https://arxiv.org/abs/2304.12210

In-class work

One person from each group will present to the class, on the board:

A decision tree for choosing an SSL strategy based on your data.

You might consider:

- What is a general framework for understanding the differences between all of these methods?
- What are some examples of these various SSL techniques applied to (text/image/physics) data?
- What questions could you ask yourself to help determine the optimal SSL strategy for a new dataset?